TITLE: Credit Risk Classification: Loan Default Prediction

Subtitle:

Using Machine Learning to Support Safer Lending Decisions

BUSINESS PROBLEM

- **Why This Problem Matters**
- Loan defaults cause significant financial losses
- Traditional credit assessment can miss high-risk borrowers
- Lenders need early and reliable risk identification
- **Business reality:**
  Approving one risky borrower can be more costly than rejecting several safe ones.

STAKEHOLDERS & GOAL

- **Who Uses This & Why**
- **Stakeholders**
- Credit risk teams at banks and fintech companies
- **Goal**
- Identify high-risk borrowers **before loan approval**
- **Outcome:**
  More informed, consistent, and safer credit decisions.

DATA OVERVIEW

- **What Data Was Used**
- Historical loan applicant data
- Includes:
  - Income and financial indicators
  - Loan characteristics
  - Applicant demographic information
  - **Target Variable**
- Whether a borrower defaulted on a loan
- **Key insight:**
  Loan defaults are imbalanced, which is common in real credit risk data.

PROJECT APPROACH

- **How the Problem Was Solved**
- Built a **classification model** to predict loan default risk
- Cleaned and prepared both numeric and categorical data
- Used a **pipeline** to ensure reproducibility
- Tested multiple models and compared performance
- **Focus:**
  Business-aligned evaluation, not just technical accuracy.

MODELS EVALUATED

- **Iterative Modeling Approach**
- Three models were tested:
- **Logistic Regression**
  - Used as a baseline for comparison
- **Decision Tree**
  - Captures non-linear patterns
- **Random Forest**
  - Ensemble model combining multiple decision trees
  - **Why multiple models?**
    To balance simplicity, performance, and business risk.

FINAL MODEL SELECTED

- **Final Model: Random Forest**
- **Why Random Forest was chosen**
- Better recall for default cases
- Handles complex borrower behavior
- Performs well on imbalanced credit data
- **Business translation:**
  It is more effective at flagging risky borrowers than simpler models.

MODEL EVALUATION

- **How Performance Was Measured**
- **Why accuracy alone is not enough**
- Misclassifying a defaulter as safe is very costly
- **Key evaluation focus**
- Recall for default cases
- Confusion matrix
- ROC-AUC score
- **Priority:**
  Catch as many high-risk borrowers as possible.

KEY RESULTS (BUSINESS VIEW)

- **What the Results Mean**
- Strong ability to identify defaulting borrowers
- Recall was prioritized to reduce risky approvals
- ROC-AUC shows good separation between defaulters and non-defaulters
- **Bottom line:**
  The model supports safer lending decisions by reducing hidden risk.

BUSINESS IMPLICATIONS

- **How This Helps the Business**
- The model can help credit risk teams to:
- Identify high-risk borrowers early
- Reduce non-performing loans
- Support risk-based pricing and approval decisions
- **mportant:**
  This model supports human decision-making — it does not replace it.

LIMITATIONS

- **What to Be Aware Of**
- Data is historical and may not reflect current economic conditions
- Informal income may not be fully captured
- Model performance depends on data quality
- **Implication:**
  The model should be monitored and retrained regularly.

CONCLUSION & RECOMMENDATIONS

- **Conclusion**
- Machine learning can effectively support credit risk assessment
- Random Forest performed best for identifying risky borrowers
- The model provides actionable, data-driven insights
- **Recommendations**
- Use as a decision-support tool
- Combine predictions with expert judgment
- Update the model with new data over time